True or false? (unless otherwise specified.) Questions 1–15: 1 point for a correct answer, 0 otherwise.

Answer grid

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	B
True	☒	☐	☒	☒	☒	☐	☒	☐	☐	☒	☒	☐	☒	☐	☒	☒
False	☐	☒	☐	☐	☐	☒	☐	☒	☒	☐	☐	☒	☐	☒	☐	☐

1. Let $x_0 < x_1 < x_2$. If $p, q \in \mathcal{P}_2$ satisfy

$$p(x_i) = y_i, \qquad q(x_i) = z_i, \qquad i = 0, 1, 2,$$

then $p + q$ interpolates the data $(x_i, y_i + z_i)$ for $i = 0, 1, 2$.

True. For $i = 0, 1, 2$, $(p + q)(x_i) = p(x_i) + q(x_i) = y_i + z_i$.

2. For all $x_0, x_1, y_0, y_1 \in \mathbb{R}$ with $x_0 \neq x_1$, there exists a unique $p \in \mathcal{P}_2$ such that

$$p(x_0) = y_0, \qquad p(x_1) = y_1.$$

False. Two points do not determine a unique polynomial in $\mathcal{P}_2$.

3. If $x_0 < \cdots < x_n$ and $\ell_i \in \mathcal{P}_n$ is the ith Lagrange basis polynomial, then by definition

$$\ell_i(x_j) = \begin{cases} 1, & i = j, \\ 0, & i \neq j, \end{cases} \qquad i, j = 0, \ldots, n.$$

True. This is the defining cardinal property of the Lagrange basis.

4. If $p, q \in \mathcal{P}_n$ and $p(x_i) = q(x_i)$ for $i = 0, \ldots, n$, where $x_0 < \cdots < x_n$, then $p = q$.

True. The difference $p - q$ has degree at most n and $n + 1$ distinct roots.

5. If $f \in \mathcal{P}_d$ and $p_n \in \mathcal{P}_n$ is its interpolant at the $n + 1$ distinct nodes $x_0, \ldots, x_n$, with $n \geqslant d$, then $p_n = f$.

True. The difference $p_n - f$ belongs to $\mathcal{P}_n$ and has $n + 1$ distinct roots.

6. **Interpolation error.** Let $f \in C^{n+3}[a, b]$, let $a \leqslant x_0 < \cdots < x_n \leqslant b$, and let $p_n \in \mathcal{P}_n$ interpolate f at $x_0, \ldots, x_n$. Then, for every $x \in [a, b]$, there exists $\xi_x \in (a, b)$ such that

$$f(x) - p_n(x) = \frac{f^{(n+3)}(\xi_x)}{(n+3)!} \prod_{i=0}^{n} (x - x_i).$$

False. The interpolation remainder uses $f^{(n+1)}(\xi_x)/(n+1)!$, not $f^{(n+3)}(\xi_x)/(n+3)!$.

7. **Interpolation error.** Let $u(x) = x^4 + 2x^3 + 3x^2 + 4$ and let $p_3 \in \mathcal{P}_3$ be its interpolant at the points $1, 2, 3, 4$. Then

$$u(x) - p_3(x) = (x - 1)(x - 2)(x - 3)(x - 4), \qquad \forall x \in \mathbb{R}.$$

True. Since $u^{(4)} = 24$, the interpolation remainder is exactly the product of the four node factors.

8. **Convergence.** For every function $f \in C[-1, 1]$, if p_n is the polynomial interpolating f at the $n + 1$ equally spaced nodes in $[-1, 1]$, including -1 and 1, then

$$\lim_{n \to \infty} \left(\max_{x \in [-1,1]} \left| p_n(x) - f(x) \right| \right) = 0.$$

False. Equidistant interpolation need not converge uniformly for every continuous function.

9. **Convergence.** Let $p_n \in \mathcal{P}_n$ be the polynomial interpolating $f(x) = \sin(2x)$ at the nodes $x_i = i/n$, for $i = 0, \ldots, n$. Then

$$\lim_{n \to \infty} \left(\max_{x \in [0,1]} \left| p_n(x) - f(x) \right| \right) = +\infty.$$

False. The function $\sin(2x)$ is analytic, and these interpolants converge uniformly on $[0, 1]$, so the limit is 0, not $+\infty$.

10. **Convergence estimate.** If $f \in C[a,b]$ and f_n is the **piecewise quadratic** interpolant of f based on a partition of $[a,b]$ into n equally large subintervals, then

$$\lim_{n\to\infty} \left(\max_{a\leqslant x\leqslant b} \left| f(x) - f_n(x) \right| \right) = 0.$$

True. Piecewise polynomial interpolation on a refining uniform mesh converges uniformly for continuous f.

11. **Chebyshev polynomials.** Fix $n \in \mathbb{N}$ and let $T_n \colon [-1,1] \to \mathbb{R}$ be defined by $T_n(x) = \cos(n \arccos x)$. Then T_n is a polynomial of degree n.

True. For $n \in \mathbb{N}$, T_n is a polynomial of degree n.

12. **Chebyshev polynomials.** Assume that T_2 is defined as in the previous item. Then $T_2(x) = 2x^2 - 3$. It may be useful to recall that $\cos(2x) = 2\cos(x)^2 - 1$.

False. The identity is $T_2(x) = 2x^2 - 1$, not $2x^2 - 3$.

13. **Chebyshev polynomials.** Let $x_0 < \cdots < x_n$ denote the roots of the Chebyshev polynomial T_{n+1} on $[-1,1]$. Using these points as interpolation nodes instead of equally spaced nodes can help reduce the maximum interpolation error.

True. Chebyshev nodes generally reduce the maximum interpolation error compared with equally spaced nodes.

14. **Julia.** Let `x = [0, 1, 2]` and `y = [1, 2, 5]`. The following Julia code uses the Gregory–Newton approach to compute the interpolating polynomial through the three data points $(0,1)$, $(1,2)$, and $(2,5)$:

```
function p(z)
    return y[1] + (y[2] - y[1]) * z
           + 1/2 * (y[3] - 2y[2] + y[1]) * z * (z + 1)
end
```

False. The Gregory–Newton factor should be $z(z-1)$, not $z(z+1)$.

15. **Julia.** Let `x = [0, 1, 2]` and `y = [1, 2, 5]`. The following Julia code correctly evaluates the Lagrange interpolating polynomial through these three data points at a point `z`:

```
function p(z)
    return (y[1] * (z - x[2]) * (z - x[3]) / ((x[1] - x[2]) * (x[1] - x[3]))
            + y[2] * (z - x[1]) * (z - x[3]) / ((x[2] - x[1]) * (x[2] - x[3]))
            + y[3] * (z - x[1]) * (z - x[2]) / ((x[3] - x[1]) * (x[3] - x[2])))
end
```

True. The three displayed terms are the Lagrange basis contributions for the three nodes.

16. **Bonus.** Let $f \in C^\infty[a,b]$, and let f_n be the **piecewise linear** interpolant of f on a uniform mesh with mesh size $h_n = \frac{b-a}{n}$. Then

$$\lim_{n\to\infty} n \times \left(\max_{x\in[a,b]} \left| f(x) - f_n(x) \right| \right) = 0.$$

Justify briefly.

True. On each mesh interval $[x_i, x_{i+1}]$, the interpolation remainder gives

$$f(x) - f_n(x) = \frac{f''(\xi_x)}{2}(x - x_i)(x - x_{i+1})$$

for some $\xi_x \in (x_i, x_{i+1})$. Since

$$\max_{x\in[x_i,x_{i+1}]} |(x - x_i)(x - x_{i+1})| = \frac{h_n^2}{4},$$

it holds that $\max_{x\in[a,b]} |f(x) - f_n(x)| \leqslant \frac{h_n^2}{8} \max_{x\in[a,b]} |f''(x)|$. Since $h_n = \frac{b-a}{n}$, the quantity in the limit is $O(n^{-1})$, so it tends to zero.